Android Practical

Module I: Android Fundamentals & UI Design

Android Basics and Resources

1) Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

<Activity.XML Code>

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:background="#FFFFFF">

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World!"
      android:textSize="30sp"
      android:textColor="#0000FF"
      android:textStyle="bold" />

</LinearLayout>
```

****

<Activity.kt>

```
package com.example.helloworld

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)
   }
}
```

****

Output:

2) Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="@dimen/activity_padding"
   android:background="@color/background_color">

   <ImageView
      android:layout_width="@dimen/image_size"
      android:layout_height="@dimen/image_size"
      android:src="@drawable/app_logo"
      android:contentDescription="@string/image_description" />

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/title"
      android:textSize="@dimen/title_size"
      android:textColor="@color/primary_color"
      android:textStyle="bold"
      android:layout_marginTop="@dimen/element_margin" />

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/description"
      android:textSize="@dimen/description_size"
      android:textColor="@color/text_color"
      android:gravity="center"
      android:layout_marginTop="@dimen/element_margin" />

   <Button
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/button_text"
      android:backgroundTint="@color/primary_color"
      android:textColor="@color/white_color"
```

```
    android:layout_marginTop="@dimen/element_margin" />

</LinearLayout>
```

****

colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <color name="primary_color">#3F51B5</color>
   <color name="background_color">#F5F5F5</color>
   <color name="text_color">#333333</color>
   <color name="white_color">#FFFFFF</color>
</resources>
```

****

strings.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <string name="app_name">ResourceDemo</string>
   <string name="title">Programming Resources</string>
   <string name="description">Welcome to Android Development!</string>
   <string name="button_text">Click Me</string>
   <string name="image_description">Application logo</string>
</resources>
```

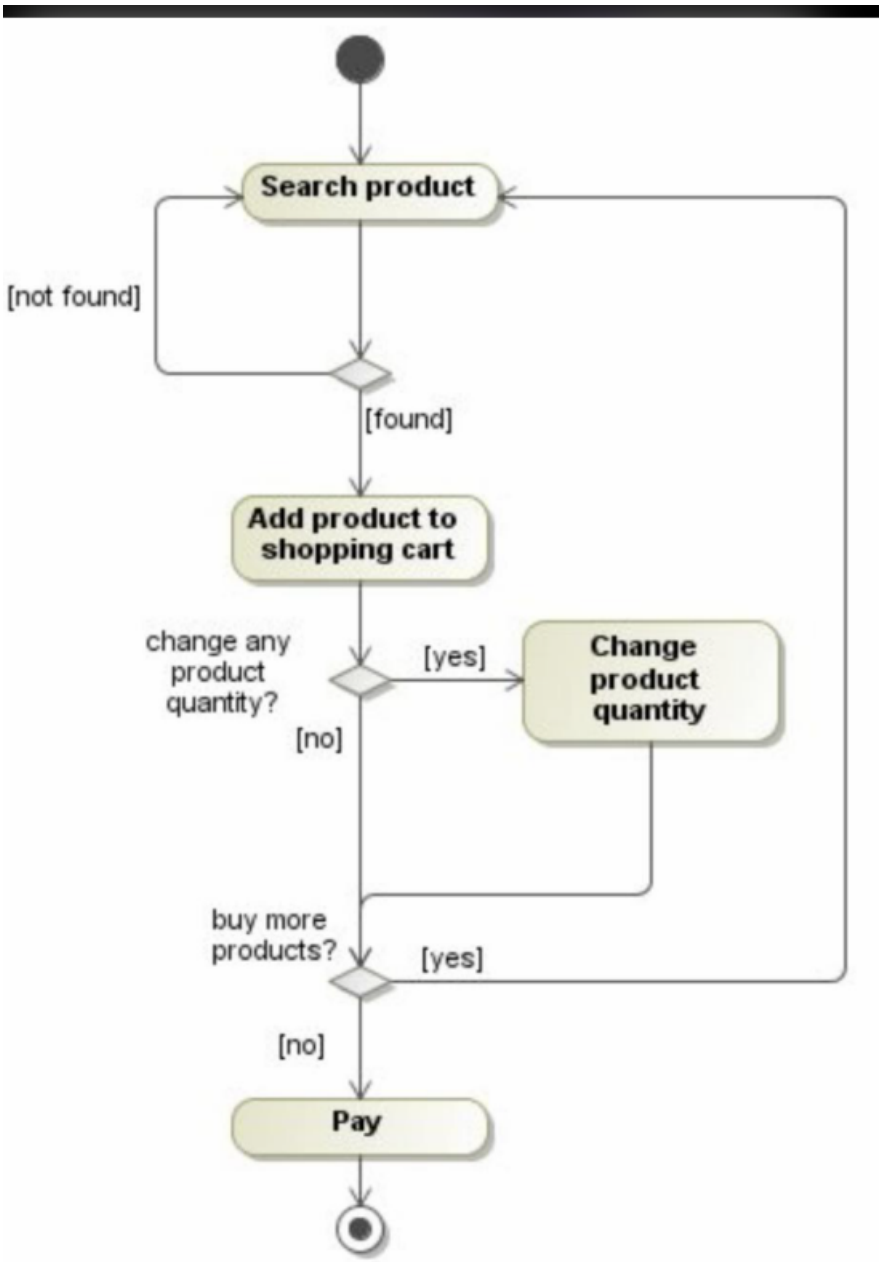
****

themes.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <style name="Theme.ResourceDemo"
      parent="Theme.Material3.DayNight.NoActionBar">
      <item name="colorPrimary">@color/primary_color</item>
      <item name="android:colorAccent">@color/primary_color</item>
      <item name="android:fontFamily">sans</item>
      <item name="android:windowLightStatusBar">true</item>
   </style>
</resources>
```

Output:

3) Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.



● onCreate(): Called by the OS when the activity is first created. This is where you initialize any UI elements or data objects. You also have the savedInstanceState of the activity that contains its previously saved state, and you can use it to recreate that state.\

fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_task_description)

● onStart(): Just before presenting the user with an activity, this method is called. It's always followed by onResume(). In here, you generally should start UI animations, audio based content or anything else that requires the activity's contents to be on screen.

●

● onResume(): As an activity enters the foreground, this method is called. Here you have a good place to restart animations, update UI elements, restart camera previews, resume audio/video playback or initialize any components that you release during onPause().
● onPause(): This method is called before sliding into the background. Here you should stop any visuals or audio associated with the activity such as UI animations, music playback or the camera. This method is followed by onResume() if the activity returns to the foreground or by onStop() if it becomes hidden.
● onStop(): This method is called right after onPause(), when the activity is no longer visible to the user, and it's a good place to save data that you want to commit to the disk. It's followed by either onRestart(), if this activity is coming back to the foreground, or onDestroy() if it's being released from memory.
● onRestart(): Called after stopping an activity, but just before starting it again. It's always followed by onStart().
● onDestroy(): This is the final callback you'll receive from the OS before the activity is destroyed. You can trigger an activity's desctruction by calling finish(), or it can be triggered by the system when the system needs to recoup memory. If your activity includes any background threads or other long-running resources, destruction could lead to a memory leak if they're not released, so you need to remember to stop these processes here as well.

Fragments and Activity Interaction

4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

fragment_login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp"
   android:background="#FFFFFF">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login Page"
      android:textSize="28sp"
      android:textStyle="bold"
      android:textColor="#3F51B5"
      android:layout_marginBottom="30dp"/>

   <EditText
      android:id="@+id/etUsername"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Username"
      android:inputType="text"
      android:padding="12dp"/>

   <EditText
      android:id="@+id/etPassword"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Password"
      android:inputType="textPassword"
      android:layout_marginTop="12dp"
      android:padding="12dp"/>

   <Button
      android:id="@+id/btnLogin"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Login"
      android:layout_marginTop="24dp"/>
```

```
</LinearLayout>
```

2.LoginFragment.kt

```
package com.example.fragmentlogin

import android.os.Bundle
import android.util.Log
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.fragment.app.Fragment

class LoginFragment : Fragment() {

    private val TAG = "LoginFragment"

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        Log.d(TAG, "onCreate called")
    }

    override fun onCreateView(
        inflater: LayoutInflater,
        container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View {
        Log.d(TAG, "onCreateView called")
        return inflater.inflate(
            R.layout.fragment_login, container, false
        )
    }

    override fun onViewCreated(
        view: View,
        savedInstanceState: Bundle?
    ) {
        super.onViewCreated(view, savedInstanceState)
        Log.d(TAG, "onViewCreated called")

        val username = view.findViewById<EditText>(R.id.etUsername)
        val password = view.findViewById<EditText>(R.id.etPassword)
        val login = view.findViewById<Button>(R.id.btnLogin)
```

```
        login.setOnClickListener {
            if (username.text.isNotBlank() &&
                password.text.isNotBlank()) {
                Toast.makeText(
                    requireContext(),
                    "Login Successful",
                    Toast.LENGTH_SHORT
                ).show()
            } else {
                Toast.makeText(
                    requireContext(),
                    "Please enter username and password",
                    Toast.LENGTH_SHORT
                ).show()
            }
        }
    }

    override fun onStart() {
        super.onStart()
        Log.d(TAG, "onStart called")
    }

    override fun onResume() {
        super.onResume()
        Log.d(TAG, "onResume called")
    }

    override fun onPause() {
        super.onPause()
        Log.d(TAG, "onPause called")
    }

    override fun onStop() {
        super.onStop()
        Log.d(TAG, "onStop called")
    }

    override fun onDestroyView() {
        super.onDestroyView()
        Log.d(TAG, "onDestroyView called")
    }

    override fun onDestroy() {
        super.onDestroy()
        Log.d(TAG, "onDestroy called")
    }
}
```

3. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/fragment_container"
   android:layout_width="match_parent"
   android:layout_height="match_parent" />
```

4. MainActivity.kt

```
package com.example.fragmentlogin

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      if (savedInstanceState == null) {
         supportFragmentManager.beginTransaction()
            .replace(R.id.fragment_container, LoginFragment())
            .commit()
      }
   }
}
```

Output:

5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.

fragment_registration.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="vertical"
      android:padding="24dp"
      android:gravity="center_horizontal">

      <TextView
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Student Registration"
         android:textSize="26sp"
         android:textStyle="bold"
         android:layout_marginBottom="24dp"/>

      <EditText
         android:id="@+id/name"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:hint="Enter Full Name"
         android:inputType="textPersonName"/>

      <EditText
         android:id="@+id/email"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:hint="Enter Email"
         android:inputType="textEmailAddress"/>

      <EditText
         android:id="@+id/mobile"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:hint="Enter Mobile Number"
         android:inputType="phone"
         android:maxLength="10"/>
```

```
        <EditText
            android:id="@+id/password"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Password"
            android:inputType="textPassword"/>

        <Button
            android:id="@+id/register"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Register"
            android:layout_marginTop="20dp"/>

        <Button
            android:id="@+id/goLogin"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Already Registered? Login"/>

    </LinearLayout>

</ScrollView>
```
Output:

6. Develop an Android application demonstrating communication between Activity and Fragment.

1. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Activity to Fragment Communication"
    android:textSize="22sp"
    android:textStyle="bold"
    android:gravity="center"
    android:padding="15dp"/>

  <EditText
    android:id="@+id/etMessage"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter your message"
    android:inputType="text"/>

  <Button
    android:id="@+id/btnSend"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send Message"/>

  <FrameLayout
    android:id="@+id/fragmentContainer"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>

</LinearLayout>
```

2. fragment_message.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fragment Screen"
        android:textSize="24sp"
        android:textStyle="bold"
        android:padding="10dp"/>

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Message will appear here"
        android:textSize="20sp"
        android:gravity="center"
        android:padding="20dp"/>

</LinearLayout>
```

Output:

Basic Layout Management

7. Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

activity_main.xml – LinearLayout (Horizontal and Vertical)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp"
   android:gravity="center_horizontal">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Student Registration"
      android:textSize="24sp"
      android:textStyle="bold"
      android:layout_marginBottom="20dp"/>

   <EditText
      android:id="@+id/etName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"
      android:inputType="textPersonName"/>

   <EditText
      android:id="@+id/etEmail"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Email"
      android:inputType="textEmailAddress"/>

   <EditText
      android:id="@+id/etMobile"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Mobile Number"
      android:inputType="phone"/>

   <!-- Horizontal LinearLayout -->

   <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
```

```
    android:orientation="horizontal"
    android:layout_marginTop="20dp">

    <Button
      android:id="@+id/btnSubmit"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="Submit"/>

    <Button
      android:id="@+id/btnCancel"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="Cancel"/>

  </LinearLayout>

</LinearLayout>
```

Output:

8. Develop an Android application using RelativeLayout to align UI components dynamically.

activity_main.xml – RelativeLayout

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="20dp">

   <TextView
      android:id="@+id/tvTitle"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Student Registration"
      android:textSize="24sp"
      android:textStyle="bold"
      android:layout_centerHorizontal="true"
      android:layout_alignParentTop="true"
      android:layout_marginTop="20dp"/>

   <EditText
      android:id="@+id/etName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"
      android:inputType="textPersonName"
      android:layout_below="@id/tvTitle"
      android:layout_marginTop="30dp"/>

   <EditText
      android:id="@+id/etEmail"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Email"
      android:inputType="textEmailAddress"
      android:layout_below="@id/etName"
      android:layout_marginTop="10dp"/>

   <EditText
      android:id="@+id/etMobile"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Mobile Number"
      android:inputType="phone"
      android:layout_below="@id/etEmail"
      android:layout_marginTop="10dp"/>
```

```
    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_below="@id/etMobile"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnCancel"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Cancel"
        android:layout_below="@id/btnSubmit"
        android:layout_marginTop="10dp"/>

</RelativeLayout>
```

Output:

9. Develop an Android application using TableLayout to design a structured Login/Registration Form.

activity_main.xml – Simple Calculator Using TableLayout

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="20dp"
   android:stretchColumns="*">

   <TableRow>
      <TextView
         android:layout_span="4"
         android:text="Simple Calculator"
         android:textSize="26sp"
         android:textStyle="bold"
         android:gravity="center"
         android:padding="20dp"/>
   </TableRow>

   <TableRow>
      <EditText
         android:id="@+id/num1"
         android:layout_span="4"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:hint="Enter First Number"
         android:inputType="numberDecimal|numberSigned"/>
   </TableRow>

   <TableRow>
      <EditText
         android:id="@+id/num2"
         android:layout_span="4"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"
         android:hint="Enter Second Number"
         android:inputType="numberDecimal|numberSigned"/>
   </TableRow>

   <TableRow>
      <Button
         android:id="@+id/add"
         android:text="+"/>
      <Button
         android:id="@+id/sub"
```

```
            android:text="-"/>
        <Button
            android:id="@+id/mul"
            android:text="×"/>
        <Button
            android:id="@+id/div"
            android:text="÷"/>
    </TableRow>

    <TableRow>
        <TextView
            android:layout_span="4"
            android:text="Result:"
            android:textSize="22sp"
            android:padding="15dp"/>
    </TableRow>

    <TableRow>
        <TextView
            android:id="@+id/result"
            android:layout_span="4"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="0"
            android:textSize="28sp"
            android:gravity="center"
            android:padding="15dp"/>
    </TableRow>

</TableLayout>
```

Output:

10) Develop an Android application using ScrollView to display long scrollable content.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="List of Animals"
            android:textSize="26sp"
            android:textStyle="bold"
            android:gravity="center"
            android:padding="10dp" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="1. Lion\n\n2. Tiger\n\n3. Elephant\n\n4. Dog\n\n5. Cat\n\n6.
Cow\n\n7. Horse\n\n8. Deer\n\n9. Monkey\n\n10. Bear\n\n11. Zebra\n\n12.
Giraffe\n\n13. Rabbit\n\n14. Fox\n\n15. Wolf\n\n16. Camel\n\n17. Sheep\n\n18.
Goat\n\n19. Kangaroo\n\n20. Panda"
            android:textSize="20sp"
            android:padding="10dp" />

    </LinearLayout>

</ScrollView>
```

Output:

11) . Develop an Android application using AbsoluteLayout demonstrating fixed positioning.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFF3E0">

    <TextView
        android:layout_width="350dp"
```

```
    android:layout_height="60dp"
    android:layout_x="10dp"
    android:layout_y="30dp"
    android:text="MEGA SHOPPING SALE"
    android:textSize="25sp"
    android:textStyle="bold"
    android:textColor="#D84315"
    android:gravity="center" />

  <TextView
    android:layout_width="300dp"
    android:layout_height="50dp"
    android:layout_x="35dp"
    android:layout_y="110dp"
    android:text="BIG DISCOUNTS!"
    android:textSize="28sp"
    android:textStyle="bold"
    android:textColor="#1565C0"
    android:gravity="center" />

  <TextView
    android:layout_width="300dp"
    android:layout_height="80dp"
    android:layout_x="35dp"
    android:layout_y="180dp"
    android:text="Clothing | Shoes | Accessories"
    android:textSize="20sp"
    android:textColor="#333333"
    android:gravity="center" />

  <TextView
    android:layout_width="300dp"
    android:layout_height="60dp"
    android:layout_x="35dp"
    android:layout_y="270dp"
    android:text="UP TO 50% OFF"
    android:textSize="30sp"
    android:textStyle="bold"
    android:textColor="#C62828"
    android:gravity="center" />

  <Button
    android:layout_width="220dp"
    android:layout_height="60dp"
    android:layout_x="75dp"
    android:layout_y="370dp"
    android:text="SHOP NOW"
    android:textSize="20sp"
```

```
        android:backgroundTint="#2E7D32"
        android:textColor="#FFFFFF" />

    <TextView
        android:layout_width="300dp"
        android:layout_height="50dp"
        android:layout_x="35dp"
        android:layout_y="460dp"
        android:text="Limited Time Offer!"
        android:textSize="20sp"
        android:textStyle="italic"
        android:textColor="#6A1B9A"
        android:gravity="center" />

</AbsoluteLayout>
```

Output:

12) . Develop an Android application using FrameLayout to overlay UI elements.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFF3E0">

    <!-- Invitation Card Background -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_margin="20dp"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="20dp"
        android:background="#FFE0B2">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="॥ श्री गणेशाय नमः ॥"
            android:textSize="24sp"
            android:textStyle="bold"
            android:textColor="#B71C1C"
            android:gravity="center"
```

```
    android:padding="10dp" />

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="GANPATI BAPPA"
    android:textSize="30sp"
    android:textStyle="bold"
    android:textColor="#E65100"
    android:gravity="center"
    android:padding="8dp" />

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="MORYA!"
    android:textSize="28sp"
    android:textStyle="bold"
    android:textColor="#BF360C"
    android:gravity="center"
    android:padding="5dp" />

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="You are Cordially Invited"
    android:textSize="20sp"
    android:textStyle="italic"
    android:textColor="#4E342E"
    android:gravity="center"
    android:padding="12dp" />

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="to celebrate the arrival of Lord Ganesha.\n\nDate: 7 September
2026\nTime: 9:30 AM\nVenue: Our Home"
    android:textSize="18sp"
    android:textColor="#3E2723"
    android:gravity="center"
    android:padding="10dp" />

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Come and seek the blessings of Bappa!"
    android:textSize="17sp"
    android:textStyle="bold"
```

```
        android:textColor="#B71C1C"
        android:gravity="center"
        android:padding="12dp" />

    </LinearLayout>

    <!-- Overlay Decorative Element -->
    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_gravity="top|center_horizontal"
      android:layout_marginTop="35dp"
      android:text="❁"
      android:textSize="35sp" />

</FrameLayout>
```

Output:

Adapter Based Views

13) Develop an Android application using ListView with ArrayAdapter and implement item click handling.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp"
  android:background="#FFFFFF">

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="List of Animals"
    android:textSize="26sp"
    android:textStyle="bold"
    android:textColor="#1565C0"
    android:gravity="center"
    android:padding="15dp" />

  <ListView
    android:id="@+id/animalList"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:divider="#BDBDBD"
    android:dividerHeight="1dp"
```

```
        android:choiceMode="singleChoice" />

    </LinearLayout>
```

Output:

14) Develop an Android application using GridView with Custom Adapter and image selection handling.

1.activity_main.xml

```
    <?xml version="1.0" encoding="utf-8"?>
    <LinearLayout
      xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:padding="12dp"
      android:background="#ECEFF1">

      <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Calculator"
        android:textSize="28sp"
        android:textStyle="bold"
        android:gravity="center"
        android:textColor="#1565C0"
        android:padding="15dp" />

      <EditText
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="90dp"
        android:hint="0"
        android:textSize="32sp"
        android:gravity="end|center_vertical"
        android:inputType="none"
        android:focusable="false"
        android:background="#FFFFFF"
        android:padding="15dp" />

      <GridView
        android:id="@+id/calculatorGrid"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:numColumns="4"
        android:horizontalSpacing="6dp"
        android:verticalSpacing="6dp"
        android:stretchMode="columnWidth"
```

```
        android:layout_marginTop="15dp"
        android:gravity="center"
        android:padding="4dp" />

</LinearLayout>
```

Output:

15) Develop an Android application implementing RecyclerView as a modern alternative to ListView.

1.activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   android:background="#FFFFFF">

   <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Student List"
      android:textSize="26sp"
      android:textStyle="bold"
      android:textColor="#1565C0"
      android:gravity="center"
      android:padding="15dp" />

   <androidx.recyclerview.widget.RecyclerView
      android:id="@+id/recyclerView"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:scrollbars="vertical" />

</LinearLayout>
```

Output: